

PE-EC702B — Digital Image and Video Processing

Semester VII • 3L:0T:0P • 3 Credits

Module 5 Exam Notes: Wavelets and Multi-resolution Image Processing

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

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1. Limitations of Fourier Transform

The Fourier Transform is highly effective at analyzing stationary signals. However, it loses all temporal information when transforming to the frequency domain.

Time-Frequency Limitations

Uncertainty Principle (Heisenberg's): States that one cannot simultaneously determine both the exact time location and frequency content of a signal with infinite precision:

$$\Delta t \cdot \Delta f \geq \frac{1}{4\pi} \quad (1)$$

where Δt is time resolution and Δf is frequency resolution.

1.1 STFT vs. Wavelet Transform

- **Short-Time Fourier Transform (STFT):** Uses a sliding temporal window $w(t)$ to localize frequencies.
 - *Limitation:* The window size is fixed. A wide window yields good frequency resolution but poor time resolution, while a narrow window yields the opposite.
- **Wavelet Transform:** Solves this by using variable window sizes (**Time-Frequency Localization**):
 - *High frequencies:* Analyzed with narrow window sizes (high time resolution, coarse frequency resolution).
 - *Low frequencies:* Analyzed with wide window sizes (coarse time resolution, high frequency resolution).

2. Continuous Wavelet Transform (CWT)

The Continuous Wavelet Transform decomposes a continuous signal into shifted and scaled versions of a prototype function $\psi(t)$ (called the **mother wavelet**).

CWT Formulations

The CWT of a signal $f(t)$ is defined as:

$$W(a, b) = \int_{-\infty}^{\infty} f(t) \psi_{a,b}^*(t) dt \quad (2)$$

where $\psi_{a,b}(t)$ is obtained by scaling and translating the mother wavelet:

$$\psi_{a,b}(t) = \frac{1}{\sqrt{|a|}} \psi\left(\frac{t-b}{a}\right) \quad (3)$$

- $a \in \mathbb{R} \setminus \{0\}$ is the **scale parameter** (inversely proportional to frequency).
- $b \in \mathbb{R}$ is the **translation parameter** (represents time shift).

- The mother wavelet must satisfy the *admissibility condition* (zero mean): $\int_{-\infty}^{\infty} \psi(t) dt = 0$.

3. Wavelet Bases and Multi-resolution Analysis (MRA)

Multi-resolution Analysis (MRA) structures the Hilbert space $L^2(\mathbb{R})$ into a nested sequence of approximation spaces V_j and detail spaces W_j .

- **Scaling Function** $\phi(t)$: Spans the approximation space V_0 . Satisfies the refinement (dilation) equation:

$$\phi(t) = \sqrt{2} \sum_n h_\phi(n) \phi(2t - n) \quad (4)$$

- **Wavelet Function** $\psi(t)$: Spans the detail space W_0 (orthogonal complement: $V_1 = V_0 \oplus W_0$). Satisfies:

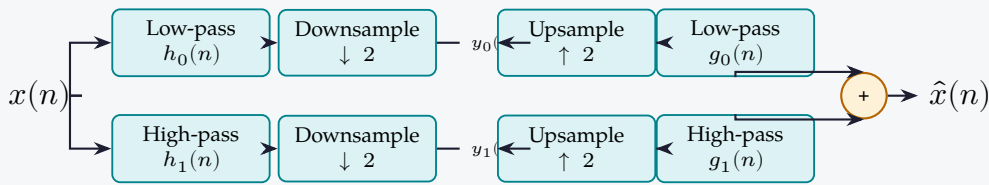
$$\psi(t) = \sqrt{2} \sum_n h_\psi(n) \phi(2t - n) \quad (5)$$

where $h_\phi(n)$ and $h_\psi(n)$ are the scaling and wavelet coefficients (filters).

4. Subband Filter Banks

Discrete wavelet decomposition and reconstruction are implemented using digital filter banks.

2-Channel Subband Coder Block Diagram



- **Analysis Bank:** Splits the signal into low-frequency approximation subband (y_0) and high-frequency detail subband (y_1). Downsampling ($\downarrow 2$) discards redundant odd samples.
- **Synthesis Bank:** Reconstructs the signal by upsampling ($\uparrow 2$, inserting zeros) and filtering.
- **Perfect Reconstruction Condition:** The filters must satisfy:

$$g_0(n) = h_0(-n), \quad g_1(n) = h_1(-n) \quad (6)$$

$$H_0(-z)G_0(z) + H_1(-z)G_1(z) = 0 \quad (\text{Alias Cancellation}) \quad (7)$$

- **Wavelet Packets:** In standard DWT, only the low-pass output is recursively decomposed. In **Wavelet Packet decomposition**, both the low-pass and high-pass branches are recursively filtered, producing a balanced tree.

5. Worked Examples

1D Haar Wavelet Transform

Problem: Given a 4-point input signal $\mathbf{x} = [4, 6, 10, 12]^T$. Apply a 1-level Discrete Haar Wavelet Transform. Use the orthogonal Haar analysis filters:

$$\text{Low-pass (approximation): } h_0 = \left[\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right]$$

$$\text{High-pass (detail): } h_1 = \left[\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}} \right]$$

Compute the approximation coefficients $\mathbf{a}$ and detail coefficients $\mathbf{d}$ after downsampling.

Solution: Discrete convolution is followed by downsampling by 2 ($\downarrow 2$). This is equivalent to applying the filters to non-overlapping pairs of input samples.

1. Calculate Approximation Coefficients (a_n):

- For the first pair $\{x(0), x(1)\} = \{4, 6\}$:

$$a_0 = \frac{1}{\sqrt{2}}x(0) + \frac{1}{\sqrt{2}}x(1) = \frac{4+6}{\sqrt{2}} = \frac{10}{\sqrt{2}} = 5\sqrt{2} \approx 7.07$$

- For the second pair $\{x(2), x(3)\} = \{10, 12\}$:

$$a_1 = \frac{1}{\sqrt{2}}x(2) + \frac{1}{\sqrt{2}}x(3) = \frac{10+12}{\sqrt{2}} = \frac{22}{\sqrt{2}} = 11\sqrt{2} \approx 15.56$$

So, the approximation coefficient vector is: $\mathbf{a} = [5\sqrt{2}, 11\sqrt{2}]^T$.

2. Calculate Detail Coefficients (d_n):

- For the first pair $\{4, 6\}$:

$$d_0 = \frac{1}{\sqrt{2}}x(0) - \frac{1}{\sqrt{2}}x(1) = \frac{4-6}{\sqrt{2}} = \frac{-2}{\sqrt{2}} = -\sqrt{2} \approx -1.414$$

- For the second pair $\{10, 12\}$:

$$d_1 = \frac{1}{\sqrt{2}}x(2) - \frac{1}{\sqrt{2}}x(3) = \frac{10-12}{\sqrt{2}} = \frac{-2}{\sqrt{2}} = -\sqrt{2} \approx -1.414$$

So, the detail coefficient vector is: $\mathbf{d} = [-\sqrt{2}, -\sqrt{2}]^T$.

Final Haar Transform Output: The reconstructed representation is the combination of approximation and detail bands:

$$\mathbf{y} = [\mathbf{a} \mid \mathbf{d}]^T = [5\sqrt{2}, 11\sqrt{2}, -\sqrt{2}, -\sqrt{2}]^T$$

Quick Revision — Module V

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Key Formulas at a Glance

Heisenberg Uncertainty	$\Delta t \cdot \Delta f \geq \frac{1}{4\pi}$
Continuous Wavelet Transform	$W(a, b) = \int_{-\infty}^{\infty} f(t) \frac{1}{\sqrt{ a }} \psi^* \left(\frac{t-b}{a} \right) dt$
Scaling refinement eq.	$\phi(t) = \sqrt{2} \sum_n h_\phi(n) \phi(2t - n)$
Wavelet refinement eq.	$\psi(t) = \sqrt{2} \sum_n h_\psi(n) \phi(2t - n)$
Haar Scaling Coefficients	$h_\phi = \left[\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right]$
Haar Wavelet Coefficients	$h_\psi = \left[\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}} \right]$

One-Line Definitions

- **Mother Wavelet:** A zero-mean, finite-energy prototype function $\psi(t)$ used to generate scaling bases.
- **MRA:** A mathematical framework that decomposes a signal into nested approximation spaces and details.
- **STFT:** A windowed Fourier transform that analyzes non-stationary signals using a fixed-width temporal slide.
- **Downsampling:** A process of discarding every alternate sample (e.g., $\downarrow 2$) to eliminate subband redundancy.
- **Wavelet Packets:** An expansion of DWT that decomposes both approximation and detail subbands recursively.
- **Admissibility Condition:** The zero-mean property required for wavelets to ensure the inverse transform exists.

Frequently Asked / Viva Questions

1. **Why is the standard Fourier Transform unsuitable for non-stationary signals?**
Because it integrates over all time ($-\infty$ to ∞), meaning it yields spectral averages but completely loses the exact time locations of transient events.
2. **Explain the trade-off in STFT window sizing.**
A wide window gives good frequency resolution but poor time resolution. A narrow window gives good time resolution but poor frequency resolution. It cannot adapt window width to frequency.
3. **How does the Wavelet Transform resolve the resolution trade-off in STFT?**
It uses variable window sizes (multi-resolution): wide windows for low-frequency details (high frequency accuracy) and narrow windows for high-frequency details.

Frequently Asked / Viva Questions (contd.)

(high time accuracy).

4. What are the physical roles of scaling and wavelet functions in MRA?

The scaling function $\phi(t)$ captures the low-frequency approximation of the signal, while the wavelet function $\psi(t)$ captures the high-frequency detail (edges).

5. What is a 2-channel subband coder?

A system of filters and samplers that splits a signal into low-frequency (approximation) and high-frequency (detail) components, downsamples them, and later reconstructs them perfectly.

6. State the perfect reconstruction (PR) condition for filter banks.

The analysis and synthesis filters must cancel aliasing caused by downsampling/upsampling ($H_0(-z)G_0(z) + H_1(-z)G_1(z) = 0$), and reconstruct the amplitude/phase without distortion.

7. What is the difference between DWT and Wavelet Packet decomposition?

DWT decomposes only the low-pass approximation subband recursively at each level. Wavelet Packet decomposition decomposes both the low-pass and high-pass subbands recursively.

8. State the properties of the Haar wavelet.

It is the oldest and simplest wavelet. It is discontinuous (step function), orthogonal, and has the shortest support (length 2), but has poor frequency selectivity.

9. Why is downsampling necessary after subband filtering?

Because filtering halves the bandwidth of the signals. Downsampling by 2 eliminates the redundant samples, keeping the total number of samples constant.

10. What is the scale parameter a in CWT?

It represents scaling/stretching: small values of a correspond to compressed wavelets (high frequencies), while large values of a correspond to stretched wavelets (low frequencies).

Fourier vs. STFT vs. Wavelet Transform

Feature	Fourier Transform	STFT	Wavelet Transform
Time Resolution	None (infinite window)	Fixed (uniform window)	Variable (multi-resolution)
Window Width	Infinite	Fixed (T)	Decreases with frequency
Suitable For	Stationary signals	Quasi-stationary	Non-stationary (transients)